Member-only story

Algorithmic Trading + Python ✓ Trading + Finance + Mean Reversion +

Mean Reversion vs. Trend Following: What 26 Years of SPY Data Actually Says

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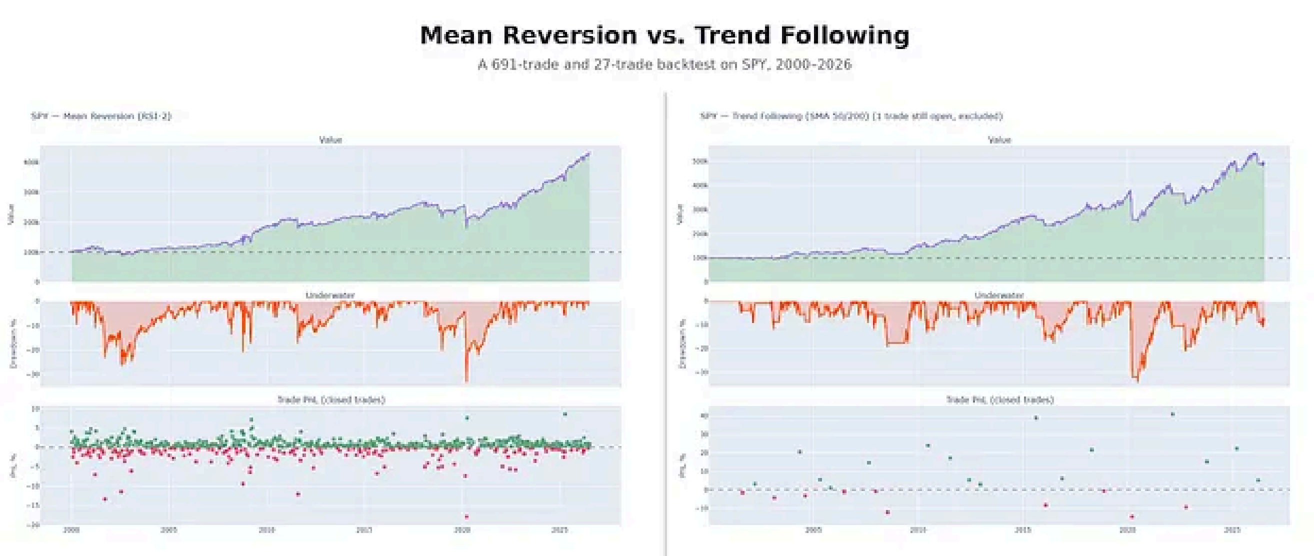


Every trading YouTube channel eventually does the "mean reversion vs. trend following" video. The pitch is always the same: mean reversion wins often but gets its face ripped off occasionally; trend following loses constantly but cashes in big when a trend finally runs. Two different P&L distributions, same theoretical expectancy, pick your poison based on personality.

It's a clean story. It's also mostly untested — delivered as a hand-drawn curve with "not to scale, no numbers on here" written right into the narration.

So I backtested it. RSI(2) mean reversion against a 50/200 SMA crossover, both long-only, both on SPY, both from January 2000 through July 2026. Here's where the narrative holds up, and where it doesn't.

The Setup

```
"""
Mean Reversion vs Trend Following — Backtest with vectorbt
-----------------------------------------------------------
Two strategies run on the same instrument/period for apples-to-apples
comparison of win rate, payoff ratio, and tail risk.

  MEAN REVERSION  : RSI(2) oversold/overbought (Connors-style)
                      BUY  when RSI(2) < RSI_OVERSOLD
                      SELL when RSI(2) > RSI_OVERBOUGHT  (or close > SMA_EXIT)

  TREND FOLLOWING : 50/200 SMA crossover (golden/death cross)
                      BUY  when SMA_FAST crosses above SMA_SLOW
                      SELL when SMA_FAST crosses below SMA_SLOW

Both are long-only for simplicity. Flip ENTRIES/EXITS logic if you want
short legs too.
"""

import pandas as pd
import numpy as np
import yfinance as yf
```

```
import vectorbt as vbt

# -------------------------
# Shared Parameters
# -------------------------
SYMBOL = "SPY"
START_DATE = "2000-01-01"
END_DATE = "2030-01-01"

INIT_CASH = 100_000
FEES = 0.0001
SLIPPAGE = 0.0002

# Mean reversion (RSI2) params
RSI_WINDOW = 2
RSI_OVERSOLD = 10
RSI_OVERBOUGHT = 70
MR_EXIT_SMA = 5          # optional: exit if close reclaims short SMA

# Trend following (SMA cross) params
SMA_FAST = 50
SMA_SLOW = 100

# -------------------------
# Download Data
# -------------------------
df = yf.download(SYMBOL, start=START_DATE, end=END_DATE, interval="1d",
                 multi_level_index=False, auto_adjust=True)
df.reset_index(inplace=True)
df["Date"] = pd.to_datetime(df["Date"])
df.set_index("Date", inplace=True)

if df.index[0] > pd.Timestamp(START_DATE):
    print(f"NOTE: {SYMBOL} data starts {df.index[0].date()}, "
          f"earlier than requested {START_DATE} (ticker inception limit).\n")

n_years = (df.index[-1] - df.index[0]).days / 365.25


def cagr_from_stats(stats, years):
    total_return_pct = stats["Total Return [%]"]
    return ((1 + total_return_pct / 100) ** (1 / years) - 1) * 100


def run_and_report(name, pf, years):
    stats = pf.stats()
    exposure_pct = pf.position_mask().mean() * 100
    cagr_pct = cagr_from_stats(stats, years)

    print("=" * 60)
```

```
    print(f"{name}: {SYMBOL}")
    print(f"Data range: {df.index[0].date()} -> {df.index[-1].date()}")
    print("=" * 60)
    print(stats)
    print("-" * 60)
    print(f"Approx. CAGR:          {cagr_pct:.2f}%")
    print(f"Market exposure:       {exposure_pct:.1f}% of trading days")
    print(f"Total trades:          {stats['Total Trades']}")
    print("=" * 60 + "\n")

    return {
        "strategy": name,
        "cagr_pct": round(cagr_pct, 2),
        "sharpe": round(stats["Sharpe Ratio"], 3),
        "sortino": round(stats.get("Sortino Ratio", np.nan), 3),
        "max_dd_pct": round(stats["Max Drawdown [%]"], 2),
        "win_rate_pct": round(stats["Win Rate [%]"], 2),
        "avg_win_pct": round(stats.get("Avg Winning Trade [%]", np.nan), 2),
        "avg_loss_pct": round(stats.get("Avg Losing Trade [%]", np.nan), 2),
        "profit_factor": round(stats.get("Profit Factor", np.nan), 2),
        "expectancy": round(stats.get("Expectancy", np.nan), 2),
        "total_trades": int(stats["Total Trades"]),
        "exposure_pct": round(exposure_pct, 1),
    }


# ==========================================================
# STRATEGY 1: Mean Reversion (RSI-2)
# ==========================================================
rsi = vbt.RSI.run(df["Close"], window=RSI_WINDOW).rsi
sma_exit = vbt.MA.run(df["Close"], window=MR_EXIT_SMA).ma

mr_entries = rsi < RSI_OVERSOLD
mr_exits = (rsi > RSI_OVERBOUGHT) | (df["Close"] > sma_exit)

pf_mr = vbt.Portfolio.from_signals(
    close=df["Close"],
    open=df["Open"],
    entries=mr_entries,
    exits=mr_exits,
    init_cash=INIT_CASH,
    fees=FEES,
    slippage=SLIPPAGE,
    freq="1d",
    direction="longonly",
)

row_mr = run_and_report("MEAN REVERSION (RSI-2)", pf_mr, n_years)

# ==========================================================
```

```
# STRATEGY 2: Trend Following (SMA Cross)
# ==========================================================
sma_fast = vbt.MA.run(df["Close"], window=SMA_FAST).ma
sma_slow = vbt.MA.run(df["Close"], window=SMA_SLOW).ma

tf_entries = sma_fast.vbt.crossed_above(sma_slow)
tf_exits = sma_fast.vbt.crossed_below(sma_slow)

pf_tf = vbt.Portfolio.from_signals(
    close=df["Close"],
    open=df["Open"],
    entries=tf_entries,
    exits=tf_exits,
    init_cash=INIT_CASH,
    fees=FEES,
    slippage=SLIPPAGE,
    freq="1d",
    direction="longonly",
)

row_tf = run_and_report("TREND FOLLOWING (SMA 50/200 Cross)", pf_tf, n_years)

# ==========================================================
# Buy & Hold benchmark
# ==========================================================
bh = vbt.Portfolio.from_holding(df["Close"], init_cash=INIT_CASH, freq="1d")
bh_stats = bh.stats()
bh_cagr = cagr_from_stats(bh_stats, n_years)

# ==========================================================
# Comparison Table
# ==========================================================
comparison_df = pd.DataFrame([row_mr, row_tf])
comparison_df.loc[len(comparison_df)] = {
    "strategy": "Buy & Hold",
    "cagr_pct": round(bh_cagr, 2),
    "sharpe": round(bh_stats["Sharpe Ratio"], 3),
    "sortino": round(bh_stats.get("Sortino Ratio", np.nan), 3),
    "max_dd_pct": round(bh_stats["Max Drawdown [%]"], 2),
    "win_rate_pct": np.nan,
    "avg_win_pct": np.nan,
    "avg_loss_pct": np.nan,
    "profit_factor": np.nan,
    "expectancy": np.nan,
    "total_trades": np.nan,
    "exposure_pct": 100.0,
}

print("\n" + "=" * 60)
print("STRATEGY COMPARISON")
```

```
print("=" * 60)
print(comparison_df.to_string(index=False))

# -------------------------
# Trade-level P&L distributions (for the article histogram)
# -------------------------
mr_trades_pnl = pf_mr.trades.records_readable["PnL"]
tf_trades_pnl = pf_tf.trades.records_readable["PnL"]

trade_pnl_df = pd.DataFrame({
    "mean_reversion_pnl": pd.Series(mr_trades_pnl.values),
    "trend_following_pnl": pd.Series(tf_trades_pnl.values),
})

trade_pnl_df.to_csv("trade_pnl_distributions.csv", index=False)
comparison_df.to_csv("strategy_comparison.csv", index=False)

print("\nSaved: trade_pnl_distributions.csv (for histogram/skew chart)")
print("Saved: strategy_comparison.csv (for the stats table)")

# -------------------------
# Skewness / kurtosis (quantifies the "fat tail" claim)
# -------------------------
print("\n" + "=" * 60)
print("DISTRIBUTION SHAPE")
print("=" * 60)
print(f"Mean Reversion  -> skew: {mr_trades_pnl.skew():.3f}, "
      f"kurtosis: {mr_trades_pnl.kurtosis():.3f}")
print(f"Trend Following -> skew: {tf_trades_pnl.skew():.3f}, "
      f"kurtosis: {tf_trades_pnl.kurtosis():.3f}")
print("(Negative skew = fatter left tail / occasional big losers,")
print(" positive skew = fatter right tail / occasional big winners)")

# -------------------------
# Plots
# -------------------------
# NOTE: pf.plot()'s built-in "trade_pnl" subplot computes marker opacity
# from trade returns, and breaks with a ValueError when any trade is
# still open at the end of the data window (open trades have NaN
# return/PnL until closed). Trend following especially tends to have
# an open trade at the end since it round-trips far less often than
# mean reversion. Rather than dropping the panel via a fallback (which
# makes the two charts inconsistent — see previous run), build the
# trade_pnl scatter manually from CLOSED trades only, on top of
# vectorbt's own value/drawdown series. This guarantees both strategies
# always get the same 3-panel layout.
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

```
def count_open_trades(pf):
    """
    vectorbt's Trades object doesn't expose a .status_open shortcut —
    'Status' (Open/Closed) lives as a column in records_readable, so
    filter on that instead.
    """
    statuses = pf.trades.records_readable["Status"]
    return int((statuses == "Open").sum())


def build_full_plot(pf, title):
    value = pf.value()
    drawdown = pf.drawdown() * 100  # to percent

    trades = pf.trades.records_readable
    closed = trades[trades["Status"] == "Closed"].copy()
    n_open = len(trades) - len(closed)

    fig = make_subplots(
        rows=3, cols=1,
        shared_xaxes=True,
        row_heights=[0.4, 0.25, 0.35],
        vertical_spacing=0.05,
        subplot_titles=("Value", "Underwater", "Trade PnL (closed trades)"),
    )

    # -- Value --
    fig.add_trace(
        go.Scatter(x=value.index, y=value.values, mode="lines", name="Value",
                   line=dict(color="mediumpurple"),
                   fill="tozeroy", fillcolor="rgba(102,187,106,0.25)"),
        row=1, col=1,
    )
    fig.add_hline(y=INIT_CASH, line=dict(color="gray", dash="dash"), row=1, col=

    # -- Underwater / Drawdown --
    fig.add_trace(
        go.Scatter(x=drawdown.index, y=drawdown.values, mode="lines", name="Draw
                   line=dict(color="orangered"),
                   fill="tozeroy", fillcolor="rgba(255,99,71,0.25)"),
        row=2, col=1,
    )

    # -- Trade PnL (closed trades only, so no NaN opacity issue) --
    if not closed.empty:
        pnl_pct = closed["Return"] * 100
        colors = ["seagreen" if p > 0 else "crimson" for p in closed["PnL"]]
        fig.add_trace(
            go.Scatter(
                x=closed["Exit Timestamp"], y=pnl_pct, mode="markers",
```

```
                marker=dict(color=colors, size=8, opacity=0.7,
                            line=dict(width=0.5, color="white")),
                name="Trade PnL %",
            ),
            row=3, col=1,
        )
    fig.add_hline(y=0, line=dict(color="gray", dash="dash"), row=3, col=1)

    subtitle = f" ({n_open} trade still open, excluded)" if n_open else ""
    fig.update_layout(title=title + subtitle, height=900, showlegend=False)
    fig.update_yaxes(title_text="Value", row=1, col=1)
    fig.update_yaxes(title_text="Drawdown %", row=2, col=1)
    fig.update_yaxes(title_text="PnL %", row=3, col=1)
    return fig


print(f"\nOpen trades at end of window -> "
      f"Mean Reversion: {count_open_trades(pf_mr)}, "
      f"Trend Following: {count_open_trades(pf_tf)}")

fig_mr = build_full_plot(pf_mr, f"{SYMBOL} — Mean Reversion (RSI-2)")
fig_mr.show()

fig_tf = build_full_plot(pf_tf, f"{SYMBOL} — Trend Following (SMA 50/200)")
fig_tf.show()
```

Mean Reversion — RSI(2): Buy when the 2-period RSI drops below 10 (oversold), exit when RSI clears 70 or price reclaims its 5-day SMA. Classic Larry Connors-style short-term mean reversion.

Trend Following — SMA 50/200 Cross: Buy on a golden cross (50-day SMA crosses above 200-day SMA), exit on a death cross. About as textbook as trend following gets.

Both run through `vectorbt` with identical cost assumptions (1bp fees, 2bp slippage) so the comparison is apples-to-apples — no strategy gets a cleaner backtest than the other.

The Distribution Shape: The Video Was Right About This Part

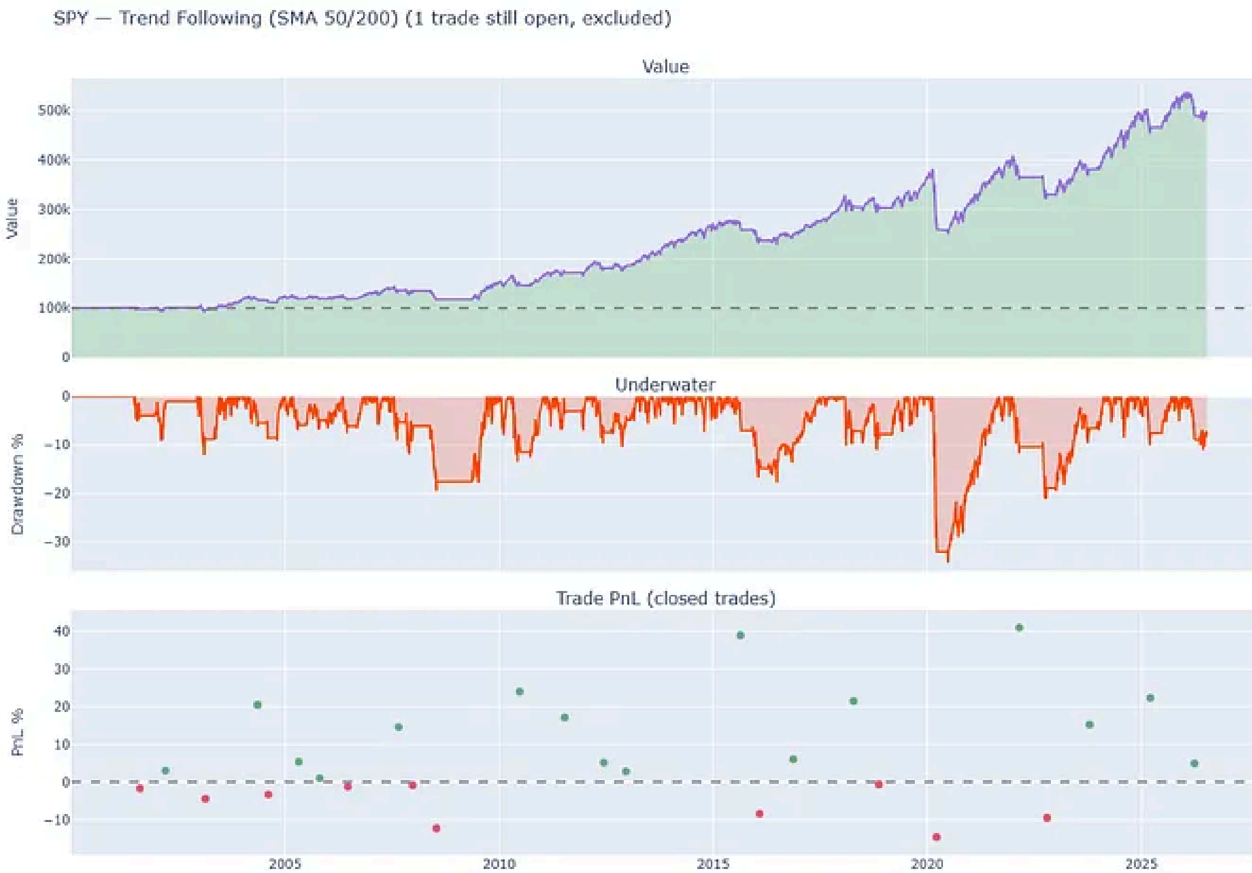
Performance Visualization

Performance Metrics

```
====================================================
MEAN REVERSION (RSI-2): SPY
Data range: 2000-01-03 -> 2026-07-14
====================================================
Start                          2000-01-03 00:00:00
End                            2026-07-14 00:00:00
Period                           6671 days 00:00:00
Start Value                                100000.0
End Value                                433836.255
Total Return [%]                         333.836255
Benchmark Return [%]                     722.064277
Max Gross Exposure [%]                        100.0
Total Fees Paid                        27997.656318
Max Drawdown [%]                          33.254431
Max Drawdown Duration            1434 days 00:00:00
Total Trades                                    691
Total Closed Trades                             691
Total Open Trades                                 0
Open Trade PnL                                  0.0
Win Rate [%]                              70.332851
Best Trade [%]                             8.513162
Worst Trade [%]                          -17.828375
Avg Winning Trade [%]                      1.077735
Avg Losing Trade [%]                      -1.767278
Avg Winning Trade Duration          1 days 23:06:40
Avg Losing Trade Duration     5 days 05:51:13.170731
Profit Factor                              1.491771
Expectancy                               483.120485
Sharpe Ratio                               0.572032
Calmar Ratio                                0.25141
Omega Ratio                                1.175225
Sortino Ratio                              0.855302
dtype: object
----------------------------------------------------
Approx. CAGR:          5.69%
Market exposure:       30.4% of trading days
Total trades:          691
====================================================
```

> 691 trades over 26 years. Notice the trade P&L panel: a dense cluster of small green wins, a scatter of red losses, and a handful of outliers stretching down to -17.8%.

Performance Visualization



Performance Metrics

```
==================================================
TREND FOLLOWING (SMA 50/200 Cross): SPY
Data range: 2000-01-03 -> 2026-07-14
==================================================
Start                          2000-01-03 00:00:00
End                            2026-07-14 00:00:00
Period                          6671 days 00:00:00
Start Value                               100000.0
End Value                            494139.239007
Total Return [%]                        394.139239
Benchmark Return [%]                    722.064277
Max Gross Exposure [%]                       100.0
Total Fees Paid                        1130.223087
Max Drawdown [%]                         34.263325
Max Drawdown Duration            544 days 00:00:00
Total Trades                                    27
Total Closed Trades                             26
Total Open Trades                                1
Open Trade PnL                                 0.0
Win Rate [%]                             61.538462
Best Trade [%]                           40.948038
Worst Trade [%]                          -14.61095
Avg Winning Trade [%]                    15.230589
Avg Losing Trade [%]                     -5.679871
Avg Winning Trade Duration       233 days 06:00:00
Avg Losing Trade Duration         74 days 16:48:00
Profit Factor                              3.95325
Expectancy                             14942.41912
Sharpe Ratio                              0.673585
Calmar Ratio                              0.266608
Omega Ratio                               1.130907
Sortino Ratio                             0.916047
dtype: object
--------------------------------------------------
Approx. CAGR:        6.21%
Market exposure:     67.8% of trading days
Total trades:        27
==================================================
```

> 27 trades over the same period. Far fewer dots, but look at the spread — two winners north of +38%, one at +40.9%.

Run the numbers on trade-level skewness and kurtosis, and the video's core claim checks out almost exactly:

```
============================================================
DISTRIBUTION SHAPE
============================================================
Mean Reversion  -> skew: -2.553, kurtosis: 27.051
Trend Following -> skew: 0.973, kurtosis: 1.081
```

```
(Negative skew = fatter left tail / occasional big losers,
positive skew = fatter right tail / occasional big winners)
```

Mean reversion's distribution is heavily left-skewed with extreme kurtosis — a tall peak of small wins sitting on top of a fat left tail. That kurtosis of 27 (versus ~3 for a normal distribution) is the "occasionally you get your face ripped off" the video described, quantified. Trend following sits on the other side: positive skew, thin tails, the shape of a strategy that loses small and wins big.

This is the one part of the YouTube narrative that survives contact with real data cleanly.

Where It Gets Interesting: Payoff Ratios

```
============================================================
STRATEGY COMPARISON
============================================================
                         strategy  cagr_pct  sharpe  sortino  max_dd_pct  win_r
            MEAN REVERSION (RSI-2)      5.69   0.572    0.855       33.25
TREND FOLLOWING (SMA 50/200 Cross)      6.21   0.674    0.916       34.26
                        Buy & Hold      8.27   0.613    0.868       55.19
```

Trend following's average win is over **13x** its counterpart in mean reversion. That's the "few big winners" story playing out exactly as advertised — a single trend trade here is worth roughly nine mean-reversion trades on average.

But notice mean reversion's payoff ratio: average win (1.08%) is actually *smaller* than the average loss (-1.77%). It's a strategy that only works because it wins nearly 70% of the time — the win rate is doing all the heavy lifting, not the size of the wins. That's a structurally fragile setup: a modest shift in market regime that drops win rate from 70% to 55% could flip this strategy negative, since the per-trade payoff math was never in its favor to begin with.

Where the Video's Narrative Breaks

Here's the claim that didn't survive: the video frames trend following as "majority of the trades are probably going to be losers… the trend will give us decent reward" — implying a low win rate offset by large winners.

The actual win rate on the SMA 50/200 crossover: **61.54%**. That's a majority of *winners*, not losers.

2 likely explanations, and they matter for how you read the rest of this table:

1. **Regime bias.** SPY from 2000–2026 spans mostly secular bull market, interrupted by three sharp drawdowns (2008, 2020, and a rocky 2022). A long-only trend-following strategy on an index that mostly goes up is going to catch more genuine uptrends than fakeouts — the video's framing describes trend following's behavior *across all markets and timeframes generally*, not one 26-year window on one large-cap US index.
2. **Sample size.** 27 trades is not a statistically robust sample. Compare that to mean reversion's 691. A single outlier trade — or the exclusion of the one trade still open at the end of the data window — can move trend following's win rate by several points. Treat every trend-following statistic here with wider error bars than the mean-reversion numbers.

The Uncomfortable Benchmark

Neither strategy beat buy-and-hold on CAGR:

- Buy & Hold: **8.27%**
- Trend Following: **6.21%**
- Mean Reversion: **5.69%**

This is the part most retail strategy content quietly skips. In a market that trends up more often than it chops, any strategy that spends time out of the market — mean reversion sits in cash 69.6% of the time, trend following 32.2% of the time — pays an opportunity cost every time it's on the sidelines during a rally. Buy-and-hold's max drawdown (55.19%) is uglier than either active strategy, which is the actual trade-off being made: both strategies bought smoother equity curves at the cost of lower absolute returns. Whether that trade is worth it depends entirely on what the capital would otherwise be doing, and on your ability to actually hold through a 55% drawdown without selling at the bottom — which is a behavioral question, not a backtest one.

The distribution shape claim holds. Skew and kurtosis numbers back up the "many small wins, rare large losses" (mean reversion) vs. "many small losses, rare large wins" (trend following) framing almost exactly as described.

The win-rate claim doesn't hold for trend following in this specific regime — a reminder that "generally speaking" claims about strategy behavior are regime-dependent, and a 26-year single-asset backtest is one draw from a much larger distribution of possible market histories.

Neither strategy justified its complexity against a passive benchmark on raw CAGR alone, on this instrument, over this period. The case for either has to be made on drawdown reduction, volatility, or how it fits inside a larger portfolio — not on "it made more money than buy-and-hold," because here it didn't.

Sample size matters as much as the math. 27 trades tells a much less reliable story than 691. Before trusting trend following's numbers on face value, they need testing across more instruments, timeframes, or regimes to see if 61.5% win rate holds up or was a product of this particular bull run.

> This article is not investment advice but is created solely for educational purposes. Investing involves risks and volatility, and users of any trading system should carefully conduct their own research before proceeding.

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